(*iv*) Ion exchange capacity of the resin to be used in the plant = 10 kg of hardness / cu m CSVTUonline.com

(*v*) Salt required for regeneration of the resin = 450 kg/cu m of the resin

The water softening plant works for 2 shifts of 8 hours per day. Assume any suitable data where necessary.

UNIT-IV

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4. (*a*) What is the function of a sluice valve? 2

(*b*) Explain the methods to remove colour, taste and odour from drinking water. 7

(*c*) Write a short note on the layout of distribution systems which are commonly used in India. 7

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(*d*) Explain the types of fire hydrants with appropriate sketches. 7

UNIT-V

5. (*a*) What are the various noise pollution control methods? 2

(*b*) Describe the causes / effects of air pollution elements viz., SO_2, CO, VOCs. 7

(*c*) Describe any two pre-cleaning devices to control air pollution (with neat sketches). 7

(*d*) Give detail use (with neat sketches) : 7

(*i*) ESP

(*ii*) Bag filters

320613(20)

BE (6th Semester)
Examination, April-May, 2016

(Old Scheme)

Environmental Engineering-I

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Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Attempt **all** five units. Part (*a*) is compulsory. Attempt any two parts from (*b*), (*c*) and (*d*) of each question.

(*ii*) The figures in the right-hand margin indicate marks.

UNIT-I

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1. (*a*) Define per capita demand. What do you understand by it, and how is it concerned with water supply engineering? 2

(*b*) Discuss the merits and demerits of river water sources and ground water source for the water supply scheme for a town. Compare the advantages and disadvantages of ground water supply and surface water supply schemes. 7

(c) From the census data given below, estimate the population of the city for the year 2030 and 2040 A.D :

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Year	1940	1950	1960	1970	1980	1990	2000	2010
Population	21610	13751	37640	46520	55460	63710	71320	79540

What are the various factors which directly affected the per capita demand of a town? What do you understand by the term 'fluctuation in water demand'?

(d) The water requirements of a city is 40×10^6 litres/day. Design a baffle wall sedimentation tank, assuming suitable design data. 7

UNIT-II

2. (a) Define the principle behind sedimentation. 2

(b) If a rectangular sedimentation tank is treating 2.5×10^6 litres/day. The size of the tank is $17.5 \times 5.5 \times 3.5$ m. If 80 p.p.m. suspended solids are present in the water, assuming the 75% removal in the basin and the average specific gravity as 2.0, determine the following : 7

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(i) Average flow of water through tank

(ii) Detention time

(iii) Deposition of the solids in the tank

(iv) Over flow rate

(c) Prove that, theoretically, the surface loading Q / A and not the depth, is a measure of effective removal of particles in a sedimentation tank. 7

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(d) Explain the significance of the following : 7

(i) Coliforms

(ii) Turbidity

(iii) Alkalinity

(iv) Chlorides

UNIT-III

3. (a) List a method NOT used for removal of taste and odour from drinking water : 2

(i) Coagulation

(ii) Pre-chlorination

(iii) Ozonization

(iv) Activated Carbon Treatment

(b) Explain the 4 minor methods of disinfection. 7

(c) Explain any two methods of removing permanent hardness in water. 7

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(d) Design a water softening plant for a water work having the following data : 7

(i) Hardness in the water = 350 mg/litre as $CaCO_3$

(ii) Quantity of water to be treated = 35000 litres/hour

(iii) Allowable hardness after treatment in the work = 75mg/litre as $CaCO_3$